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RESEARCH

Er:YAG and Nd:YAG-based low-level laser therapy (LLLT) with medical collagen improve third-molar extraction wound healing: a randomized controlled trial

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Abstract

To evaluate whether Er: YAG surgery and Nd: YAG-based low-level laser therapy (LLLT), combined with medical collagen, improve early postoperative outcomes after extraction of impacted mandibular third molars. Adults undergoing surgical removal of mesioangular or horizontal impacted mandibular third molars were randomized equally (1:1:1:1) to blank control (BCG), collagen only (COL), lasers only (LAS), or collagen plus lasers (COL+LAS). **Primary endpoints** were pain (VAS; spontaneous and swallowing; POD1/3/7), swelling (POD1/3/7), trismus (POD1/3/7), and early mucosal healing (POD3/7). **Secondary endpoints** were bleeding (30 min post-extraction only), cutaneous induration/ecchymosis (POD7), alveolar osteitis (POD3/7), and local cytokines (IL-1β, IL-6, TNF-α; POD3/7). In 120 patients, **COL+LAS** was superior: less pain, swelling, and trismus at **POD1/3** and higher early mucosal-healing rates at **POD3/7** than **BCG**, **COL**, or **LAS**; by **POD7**, between-group differences in pain, swelling, and trismus were minimal. All active arms reduced **30-min bleeding** versus BCG. At **POD7**, cutaneous induration was lowest with COL+LAS (ecchymosis not significant). Cytokines declined overall; **IL-1β** and **TNF-α** were lowest with COL+LAS at **POD3/7**, whereas **IL-6** decreased similarly across active arms. Er: YAG surgery plus Nd: YAG-based LLLT with medical collagen improved short-term recovery after impacted mandibular third-molar surgery.

Keywords Impacted teeth · Er:YAG laser · Nd:YAG laser · Low-level laser therapy · Collagen · Postoperative complications

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Introduction

Impacted teeth are teeth that fail to erupt completely or at all because of resistance from soft tissues, bone, or adjacent teeth; the condition most commonly involves maxillary and mandibular third molars [1]. Third molars are linked to dental caries, pericoronitis, damage to adjacent teeth and periodontal tissues, and an increased risk of cysts and tumors [2].Whether asymptomatic third molars should be removed prophylactically remains debated; current recommendations favor individualized risk–benefit evaluation [3]. In clinical practice, mandibular third-molar extraction is among the most frequent dentoalveolar procedures [4]. Nevertheless, postoperative pain, bleeding, swelling, trismus, and impaired wound healing remain common [5]. As living standards improve, patient expectations for minimally invasive, painless extraction continue to increase. Therefore, safe and effective strategies are needed to minimize adverse events after extraction of impacted teeth [6].

In recent years, several adjunctive materials for filling extraction sockets have demonstrated efficacy in reducing complications, including hyaluronic acid [7], medical collagen [7, 8], platelet-rich fibrin (PRF) [9], and guided tissue regeneration using resorbable or non-resorbable membranes with bone substitutes [10]. Recent clinical reports indicate that oral probiotics can modulate the oral microbiome and inflammatory milieu, thereby improving wound healing and postoperative comfort after oral surgery [11, 12].Medical collagen—typically produced by low-temperature lyophilization of bovine Achilles-tendon type I collagen, which is homologous to human type I collagen—is widely used for tissue repair. As a scaffold, collagen forms a stable, cross-linked network that supports cell migration and proliferation, thereby promoting tissue growth and wound healing. Owing to its ability to dampen postoperative inflammation, shorten the duration of post-extraction pain, lessen trismus, and accelerate alveolar-bone remineralization [13], medical collagen has become a commonly used adjunct for socket management in oral and maxillofacial surgery [7].

The Er: YAG laser (2,940 nm) is strongly absorbed by water and hydroxyapatite, and ablates tissue via a water-mediated, thermomechanical process in which rapid vaporization produces micro-explosions [14]. Consequently, compared with Nd: YAG and diode lasers, Er: YAG irradiation minimizes collateral thermal effects at the tissue surface. This mechanism—often described as "cold ablation"—reduces thermal injury to alveolar bone relative to conventional turbines [14]. The Er: YAG laser also exerts photobiomodulatory effects, promoting type I collagen formation and osteogenesis [15]. In addition, Er: YAG irradiation provides biostimulatory and antimicrobial actions on wound surfaces [16].

The Nd: YAG laser (1,064 nm) penetrates deeply into tissue and is therefore of particular interest for promoting healing. Its primary mechanism is melanin-mediated absorption of laser energy, which generates heat and produces a photothermal effect [17]. Pigmented bacteria, such as Porphyromonas gingivalis, are susceptible to this thermal action, yielding bactericidal effects [18]. The Nd: YAG laser also inactivates bacterial virulence factors, including endotoxin (lipopolysaccharide), thereby reducing the risk of postoperative infection [19]. Owing to its penetration depth, Nd: YAG irradiation can induce cellular coagulation—including of neural elements—which decreases nociceptor sensitivity and contributes to hemostasis and analgesia [20].

This study evaluates the therapeutic efficacy and clinical feasibility of combining Er: YAG and Nd: YAG lasers with medical collagen for post-extraction management of sockets after impacted third-molar removal. Our objective is to optimize minimally invasive extraction workflows, reduce postoperative complications, and improve patient comfort.

Materials and methods

Ethics and registration

The study was approved by the Ethics Committee of the Hospital of Stomatology, Zhongshan City (Approval No. [2024] Research Project No. 7; March 18, 2024) and conducted in accordance with the Declaration of Helsinki. The trial was registered with the Chinese Clinical Trial Registry (PID 248812). Written informed consent was obtained from all participants.

Trial design and setting

Prospective, single-center, four-arm, parallel-group, superiority randomized controlled trial with equal allocation (1:1:1:1), conducted from October 2023 to June 2024 and reported in accordance with CONSORT. Only one index mandibular third molar per participant was included (parallel design; no split-mouth).

Participants (Eligibility Criteria)

Inclusion Adults scheduled for removal of a mandibular third molar with mesioangular or horizontal impaction (Pell–Gregory class B/C) requiring mucoperiosteal flap elevation; no acute infection; medically fit; able to comply with follow-up.

Exclusion Uncontrolled systemic disease affecting perioperative risk (e.g., hypertension, heart disease, diabetes, hyperthyroidism, hematologic disorders); pregnancy or menstruation; acute pericoronitis/infection within 7 days; use within 7 days of anticoagulants/antiplatelets, hemostatics, psychotropics, analgesics, or oral contraceptives;

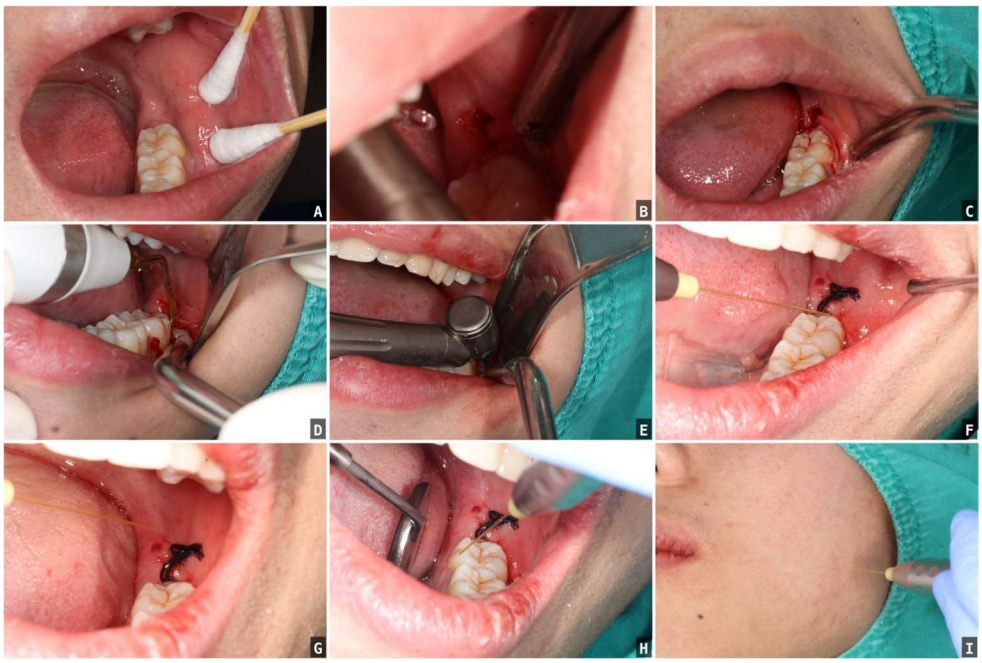


Fig. 1 Surgical workflow for impacted mandibular third-molar extraction with adjunctive lasers and collagen. **A**. Pre-operative photograph of the operative site. **B**-**C**.Er: YAG laser gingival incision. **D**. Piezoelectric osteotomy (PIEZOTOME) for conservative bone removal. **E**. Tooth sectioning with a high-speed, motor-driven electric contra-angle handpiece. F. Nd: YAG low-level laser photobiomodulation applied to the operative site. **G**. Photobiomodulation to anterior tonsillar pillar. **H**. Photobiomodulation to the pterygomandibular region. **I**. Photobiomodulation to the masseter region overlying the operative side

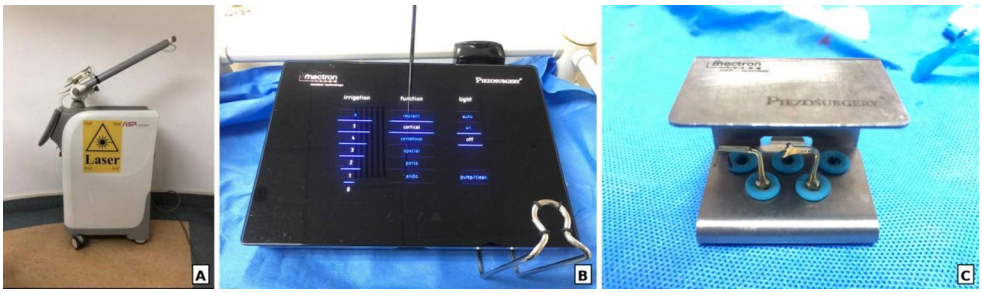


Fig. 2 Dual-wavelength laser and ultrasonic osteotomy systems. **A**, Er: YAG & Nd: YAG laser unit (Fotona Light Walker). **B**, Piezoelectric bone surgery console (Mectron Piezosurgery). **C**, Piezoelectric bone surgery handpieces and tips set

collagen allergy; jaw cysts/tumors; periodontitis with bleeding on probing or probing depth > 3 mm and/or attachment loss; other conditions affecting hemostasis or wound healing.

Interventions

Randomization was generated pre-incision using a computer-based random-number table and implemented by personnel independent of the surgeons via sequentially numbered, opaque, sealed envelopes. Participants were assigned in equal numbers ($n = 30$/arm) to BCG, COL, LAS, or COL + LAS.

Surgical Procedure(Figs. 1 and 2)

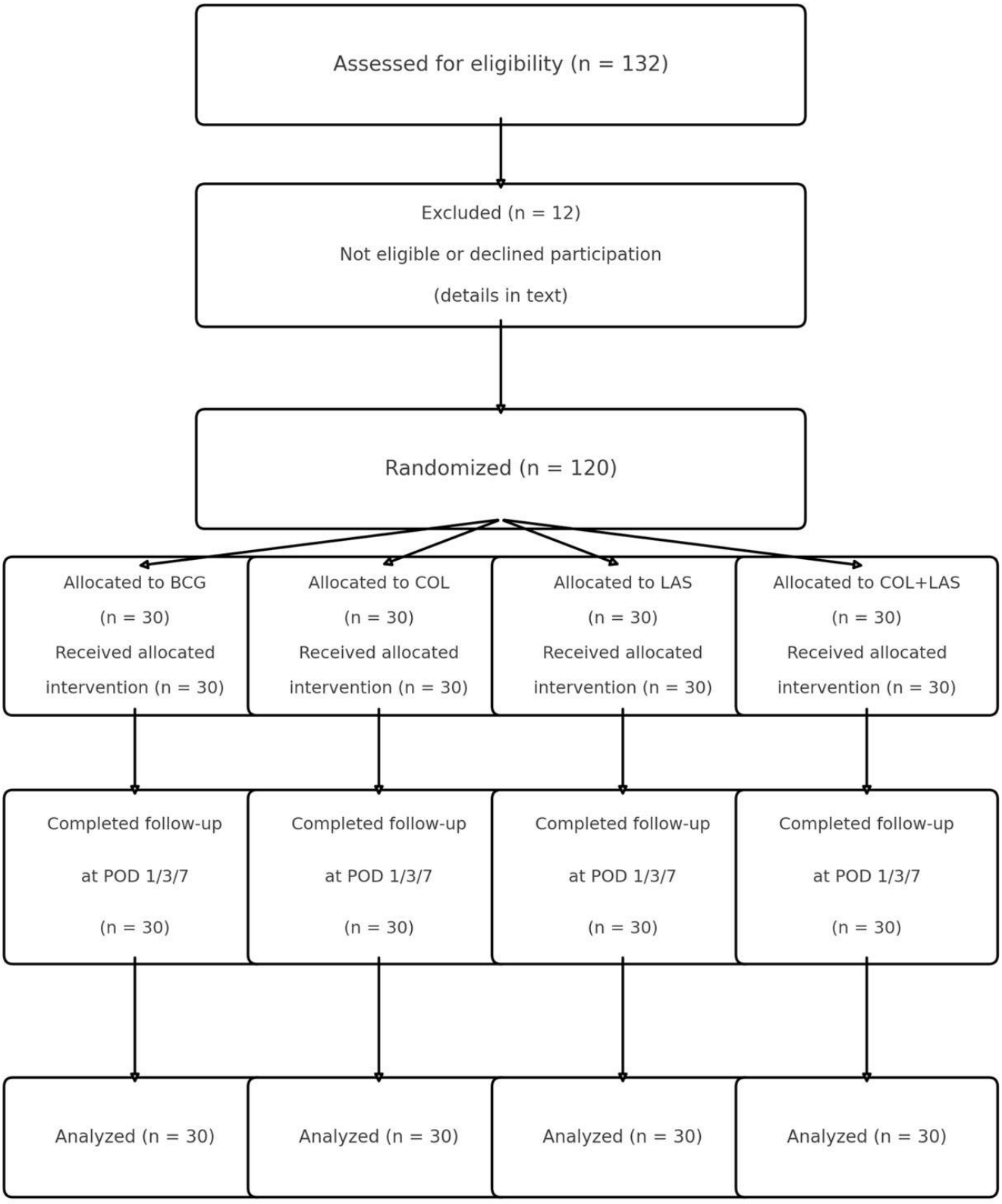
(1) Asepsis: 0.2% chlorhexidine rinse, perioral 7.5% povidone–iodine, sterile draping.
(2) Anesthesia: inferior alveolar, lingual, and buccal nerve blocks with articaine/epinephrine.
(3) Incision/flap: oblique mesiobuccal incision at the second molar with a distal relieving incision at the third; full-thickness mucoperiosteal flap.
(4) Ostectomy: PIEZOTOME (US1) to expose the maximal tooth circumference.
(5) Space creation: UC1/UC5 tips.
(6) Sectioning/delivery: 45° high-speed contra-angle handpiece and elevators. All procedures were performed by a single surgeon.

BCG (blank control): socket curettage and suturing; sham laser (device off with identical handling and sound cues).

COL (collagen only): as BCG plus placement of a collagen cylinder (7 × 15 mm) in the socket, then suturing.

LAS (lasers only):

- Er: YAG—incision: 1.0-mm fiber, SP, 120 mJ, 15 Hz (2.4 W); water 4, air 4. Gingival separation: 0.5-mm fiber, MSP, 40 mJ, 20 Hz; water 6, air 4.
- Er: YAG—socket degranulation: MSP, 80 mJ, 20 Hz (1.6 W); water 8, air 4; sweep all socket walls for 5 min.
- Nd: YAG LLLT: MSP, 15 Hz, 1.5 W; handpiece ~ 1 cm from the gingiva; sites: distal-incision center, mesial, distal, buccal, lingual; mesial incision; anterior tonsillar



CONSORT flow diagram for a four-arm randomized controlled trial.
BCG: blank control; COL: collagen; LAS: lasers; COL+LAS: collagen plus lasers.

Fig. 3 CONSORT participant flow diagram

Table 1 Baseline characteristics of participants by treatment group

Group	BCG	COL	LAS	COL+LAS	Total	*P*value
Age (years)	28.00±6.48	29.23±7.48	31.03±7.27	29.60±6.57	29.47±6.96	0.977
Sex						
Male	14	15	16	13	58	0.881
Female	16	15	14	17	62	
Position						
Left	14	17	15	13	59	0.761
Right	16	13	15	17	61	
Type of Obstruction						
Median obstruction	15	15	15	15	60	1
Low obstruction	15	15	15	15	60	
	30	30	30	30	120	

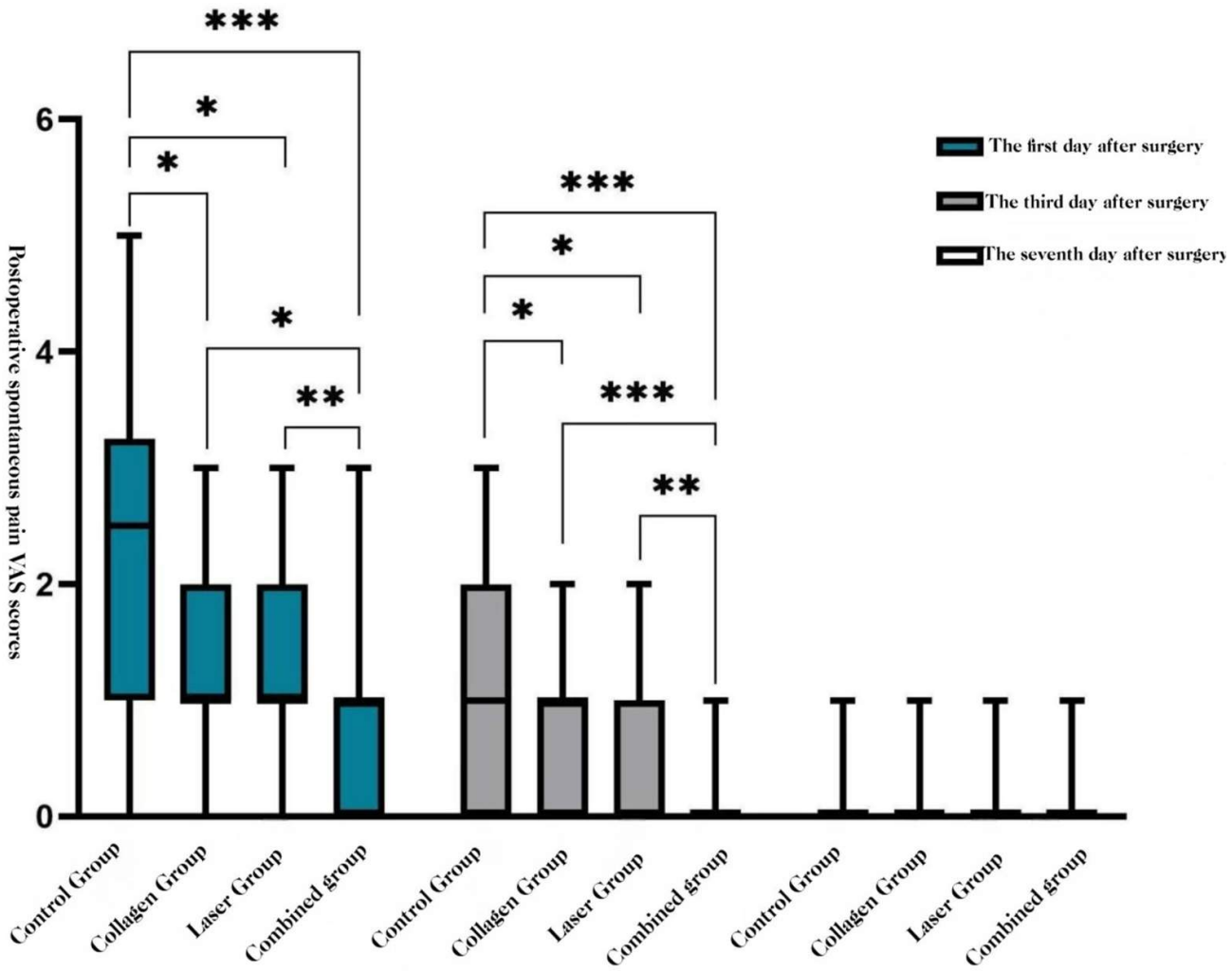


Fig. 4 Spontaneous pain VAS by group (POD 1, 3, 7): box-and-whisker plots.

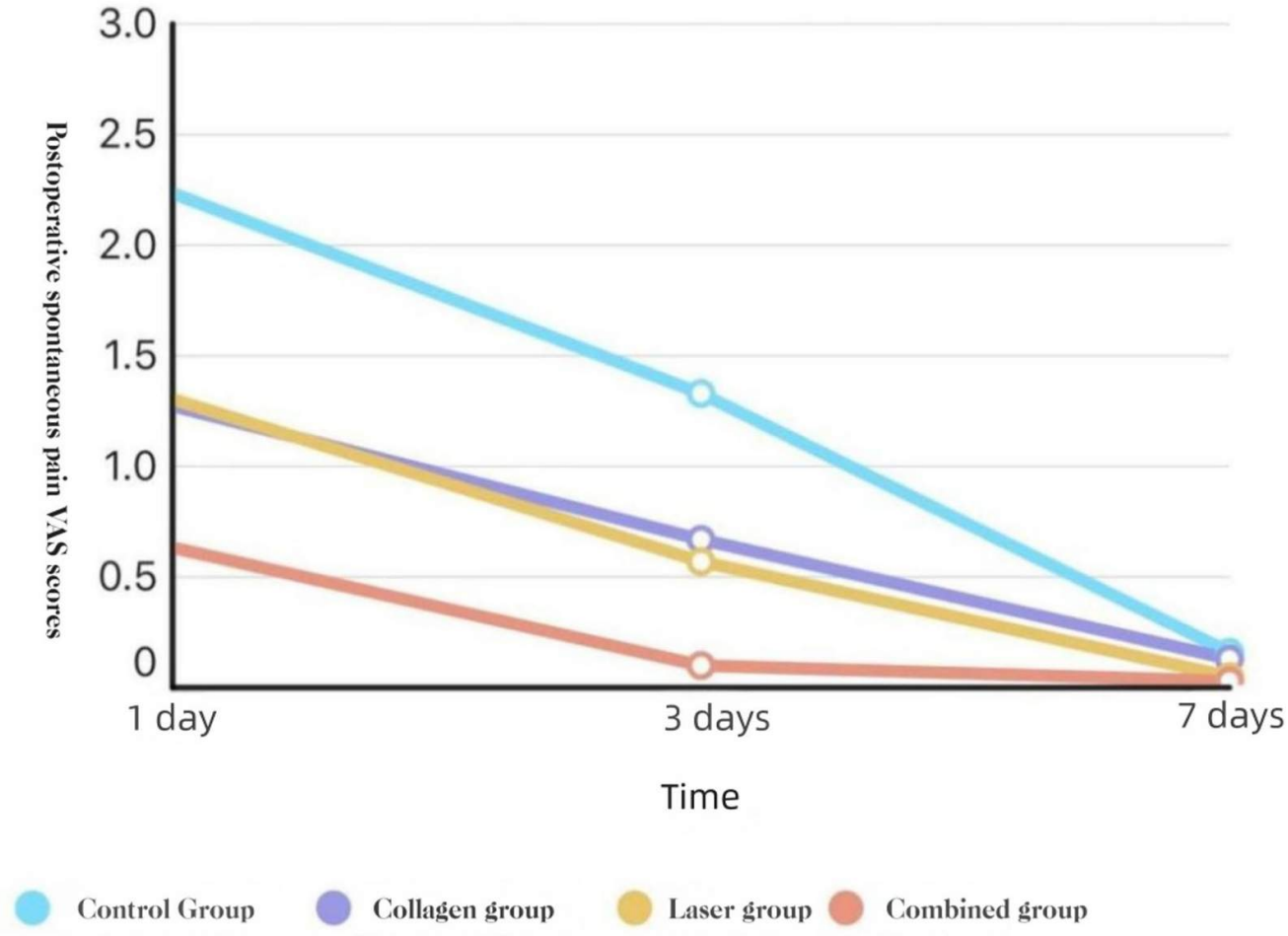


Fig. 5 Spontaneous pain VAS over time by group (POD 1, 3, 7):line graph of group means

pillar; pterygomandibular region; masseter; ~5 min per site; flap sutured.

COL+LAS (collagen plus lasers): identical Er: YAG steps and Nd: YAG LLLT as above, then collagen cylinder placement and suturing.

Outcomes and assessments

Primary endpoints

Pain (VAS 0–10; POD1/3/7): two domains recorded separately on a 10-cm VAS (0 = no pain; 10 = worst imaginable): spontaneous pain at rest and swallowing-evoked pain during deglutition (standardized instructions).

Swelling (POD1/3/7): Marković–Todorović [21] linear distance (from the midline-inferior chin point to the junction of the earlobe's lower border with cheek skin), measured three times and averaged; grading by change from baseline: Grade I < 2 mm, Grade II 2–4 mm, Grade III >4 mm.

Trismus (POD1/3/7): maximum interincisal distance (mm) measured with a caliper (×3, averaged) and reported as change from baseline (ΔMID).

Early mucosal healing [22, 23] (POD3 and POD7; 3-grade rubric):

- POD3 — Good: pink/normal margin, socket contraction, no edema, free opening; Fair: mild marginal erythema limited to the wound edge with slight discomfort, modest granulation; Poor: deep erythema extending to attached gingiva/adjacent tooth, edema or clot breakdown/absent granulation, mouth-opening discomfort.
- POD7 — Good: complete epithelial coverage, no erythema/edema, granulation fills the socket; Fair: slight residual erythema with near-complete granulation fill;

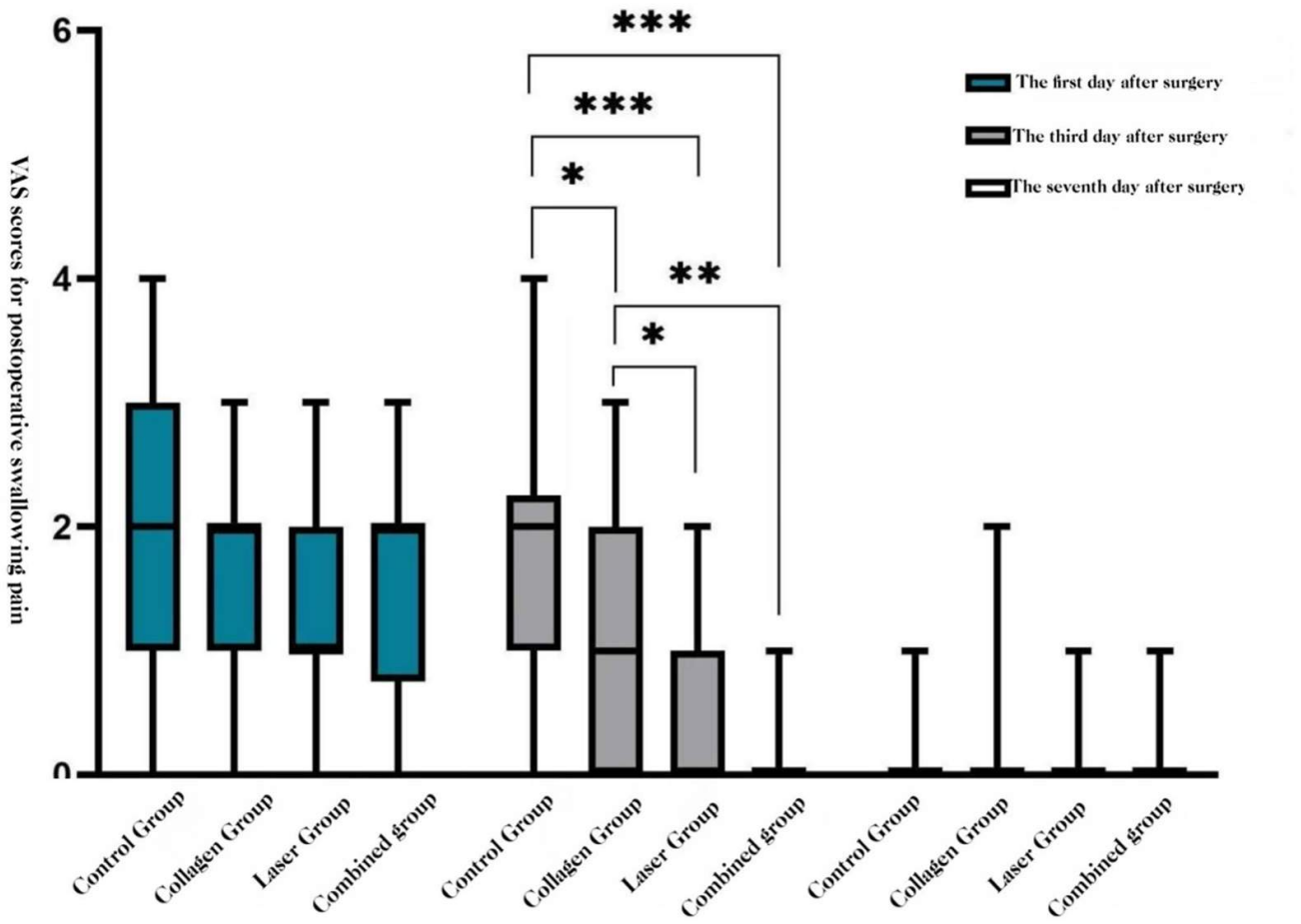


Fig. 6 Swallowing-evoke VAS by group (POD 1, 3, 7): box-and-whisker plots

Poor: deep erythema with marginal/attached-gingival swelling, scant granulation, partially empty socket.

Secondary endpoints

Bleeding (30 min post-op only; 3-level): 0 = none (gauze minimally stained, no active socket bleeding); 1 = mild (gauze largely stained, blood-streaked saliva, no active socket bleeding); 2 = marked (gauze fully soaked with active oozing and/or visible clots).

Cutaneous induration/ecchymosis (POD7): presence/absence of cheek ecchymosis or palpable induration at mandibular border/buccal tissues.

Alveolar osteitis (POD3 and POD7): present/absent per standard criteria—severe pain peaking 48–72 h post-extraction with fetor and an empty or disintegrating clot.

Local cytokines (POD3 and POD7): socket exudate collected after isolation and air-drying: four paper points placed for 30 s (mesiobuccal and mesiolingual aspects), pooled into EP tubes and stored at −80 °C; eluates (PBS, 10,000 rpm, 15 min) assayed by ELISA for IL-1β, IL-6, and TNF-α per manufacturer instructions.

Sample size determination

Pilot day-3 rates (Oct 2023–Feb 2024) for spontaneous pain, swelling, and trismus informed power. With two-sided $\alpha = 0.05$ and 90% power (PASS 2023), required totals for the four-arm design were 76, 72, and 84, respectively. Allowing 20% attrition, the largest requirement yielded $N = 104$ (26/group); the target enrollment was $N = 120$.

Blinding

The BCG group received a sham laser (device off with identical handling/sound cues). For safety, surgeons were aware

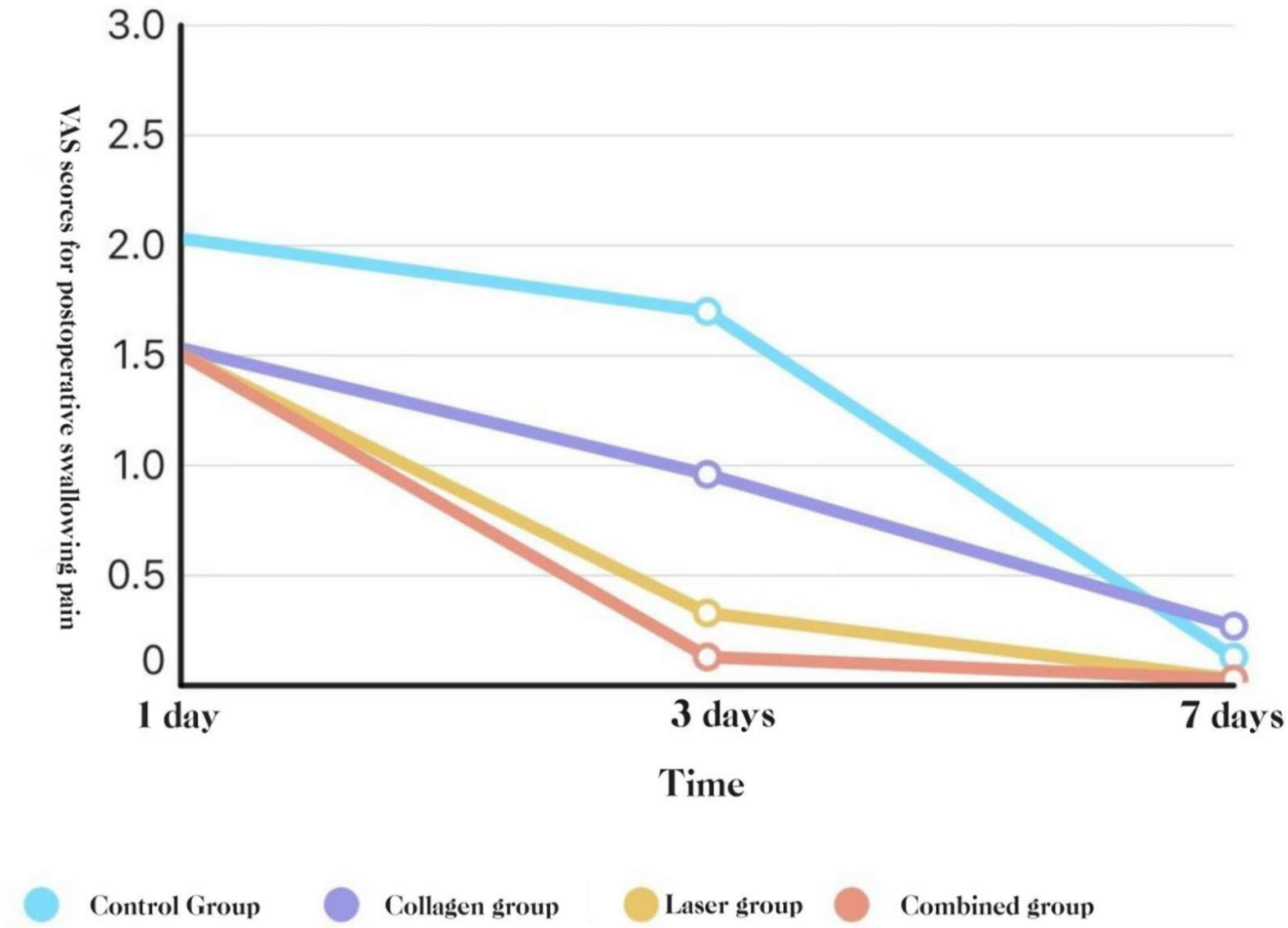


Fig. 7 Swallowing-evoke VAS over time by group (POD 1, 3, 7): line graph of group means. * P < 0.05; ** P < 0.01; *** P < 0.001

of energy delivery; participants, outcome assessors, and data analysts were blinded.

Statistical analysis

Normality (Shapiro–Wilk) and homogeneity of variance (Levene) were assessed. Continuous outcomes were analyzed with linear mixed-effects models (fixed: group, time, group×time; random: subject intercept). Ordinal outcomes used proportional-odds models; binary outcomes used χ^2 or Fisher's exact tests. Multiplicity was controlled with Tukey (parametric) or Holm–Bonferroni/Dunn procedures, as appropriate. Two-sided $P < 0.05$ was considered significant. Analyses were performed in SPSS v27.0.

Results

Participant flow Of 132 patients screened, 120 were randomized and included in the analysis (BCG $n = 30$; COL $n = 30$; LAS $n = 30$; COL + LAS $n = 30$) **(**Fig. 3**)**.

Baseline characteristics Groups were comparable for age, sex, side, and impaction depth (all $P > 0.05$) **(**Table 1**)**.

Primary endpoints

Pain (VAS)

Spontaneous pain At POD 1 and POD 3, COL + LAS was lower than all other groups, and COL and LAS were each

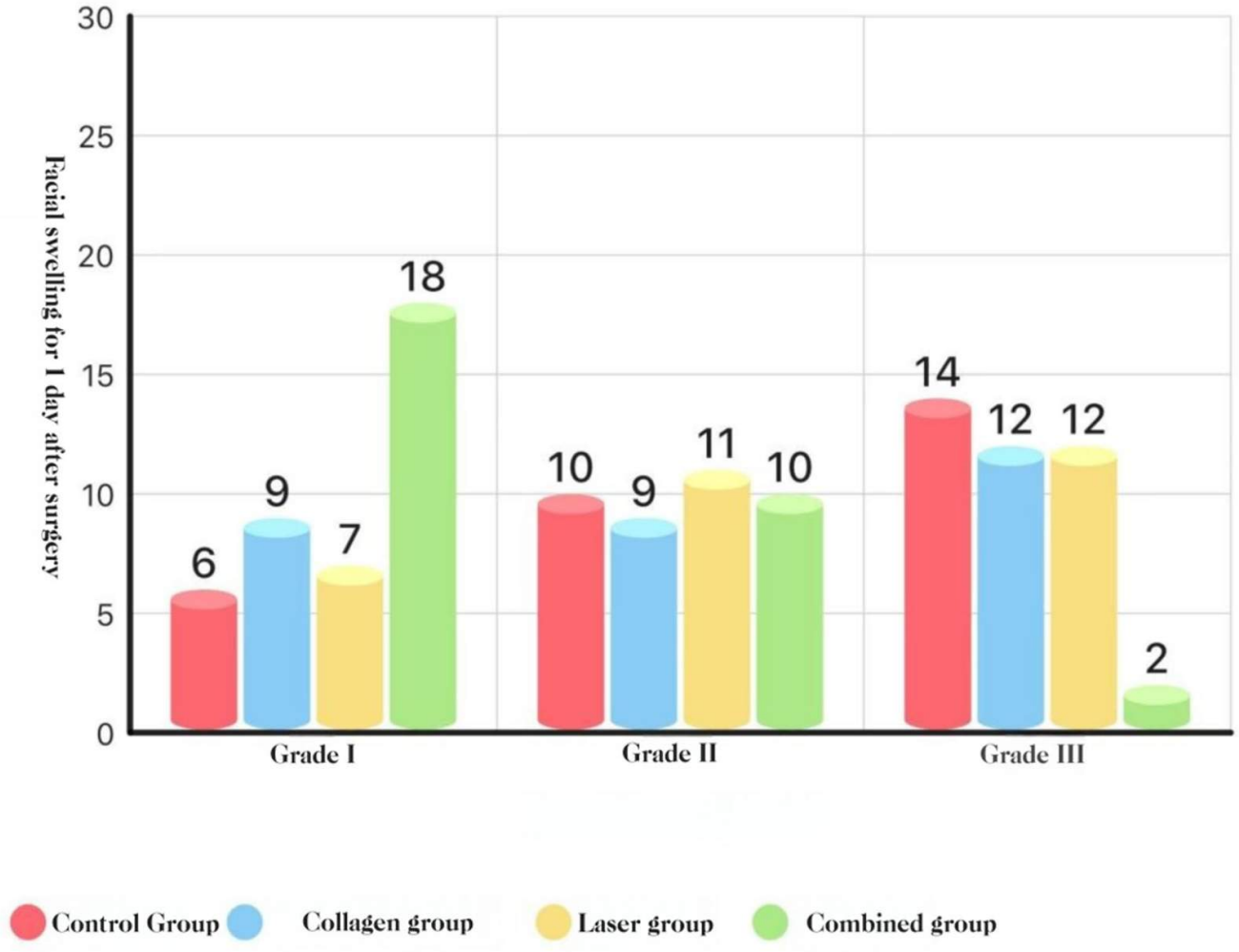


Fig. 8 Distribution of facial swelling at POD 1 (Marković–Todorović grades): histogram

lower than BCG (all $P<0.05$). At POD 7, no group differences were observed ($P>0.05$).**See** Figs. 4 and 5.

Swallowing-evoked pain At POD 3, COL+LAS and LAS were lower than COL and BCG, and COL was lower than BCG (all $P<0.05$). At POD 1 and POD 7, no group differences were observed ($P>0.05$).**See** Figs. ,6, and 7

Swelling (graded)

POD 1: COL+LAS was lower than COL, LAS, and BCG ($P<0.05$).

POD 3: COL+LAS was lower than all other groups, and COL and LAS were each lower than BCG (all $P<0.05$).

POD 7: no group differences ($P>0.05$).**See** Figs. 8,9, and 10.

Trismus (change in maximum interincisal distance).

POD 1: COL+LAS was lower than BCG, COL, and LAS ($P<0.05$).

POD 3: COL+LAS and LAS were lower than COL and BCG ($P<0.05$).

POD 7: no group differences ($P>0.05$).See Figs. 11 and 12.

Early mucosal healing (three-grade rubric)

At POD 3 and POD 7, COL+LAS achieved better healing than all other groups, and COL and LAS each outperformed BCG (all $P<0.05$).See Figs. 13 and 14.

Secondary endpoints

Bleeding at 30 min (three-level scale)

All active arms (COL, LAS, COL+LAS) were lower than BCG ($P<0.05$), with no differences among the active arms ($P>0.05$).**See** Fig. 15.

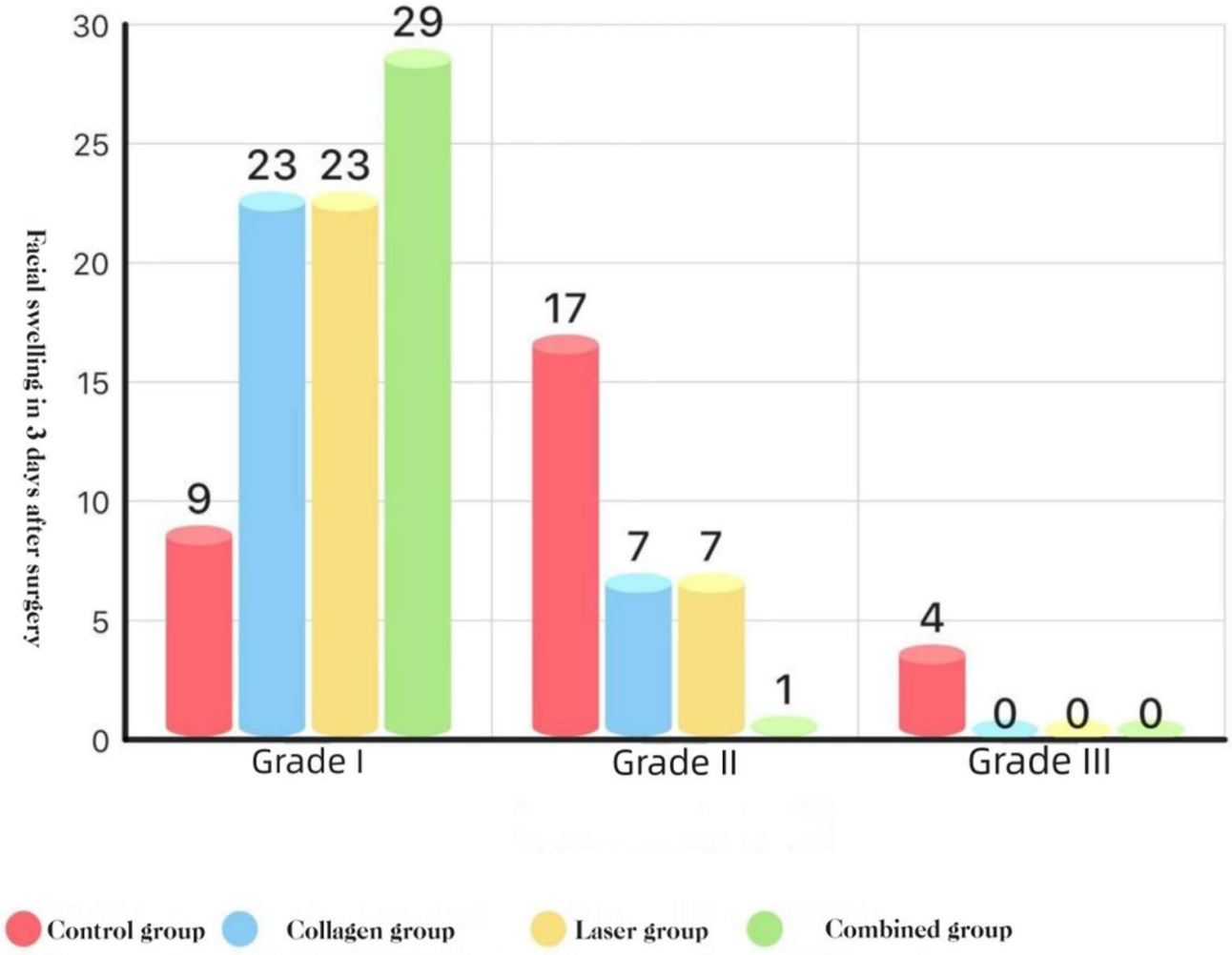


Fig. 9 Distribution of facial swelling at POD 3(Marković–Todorović grades): histogram

Cutaneous induration and ecchymosis (POD 7)

Induration: COL+LAS was lower than the other three groups, and COL and LAS were each lower than BCG (all $P<0.05$).

Ecchymosis: no group differences ($P>0.05$).**See** Fig. 16, and 17.

Alveolar osteitis

None occurred in any group.

Local cytokines (socket exudate, ELISA)

IL-1β. POD 3: COL+LAS was lower than BCG and LAS; COL was lower than BCG ($^{P}<0.05$). POD 7: COL+LAS was lower than all other groups, and COL and LAS were each lower than BCG ($P<0.05$).

TNF-α. POD 3: COL+LAS was lower than the other three groups ($P<0.05$). POD 7: COL+LAS was lower than all other groups, and COL and LAS were each lower than BCG ($P<0.05$).

IL-6. POD 3: COL+LAS was lower than all other groups, and COL and LAS were each lower than BCG ($P<0.05$). POD 7: COL, LAS, and COL+LAS were all lower than BCG ($P<0.05$), with no differences among the active arms ($P>0.05$).**See** Fig. 18, 19 and 20.

COL + LAS: postoperative facial swelling and mucosal healing

(Figure 21.)

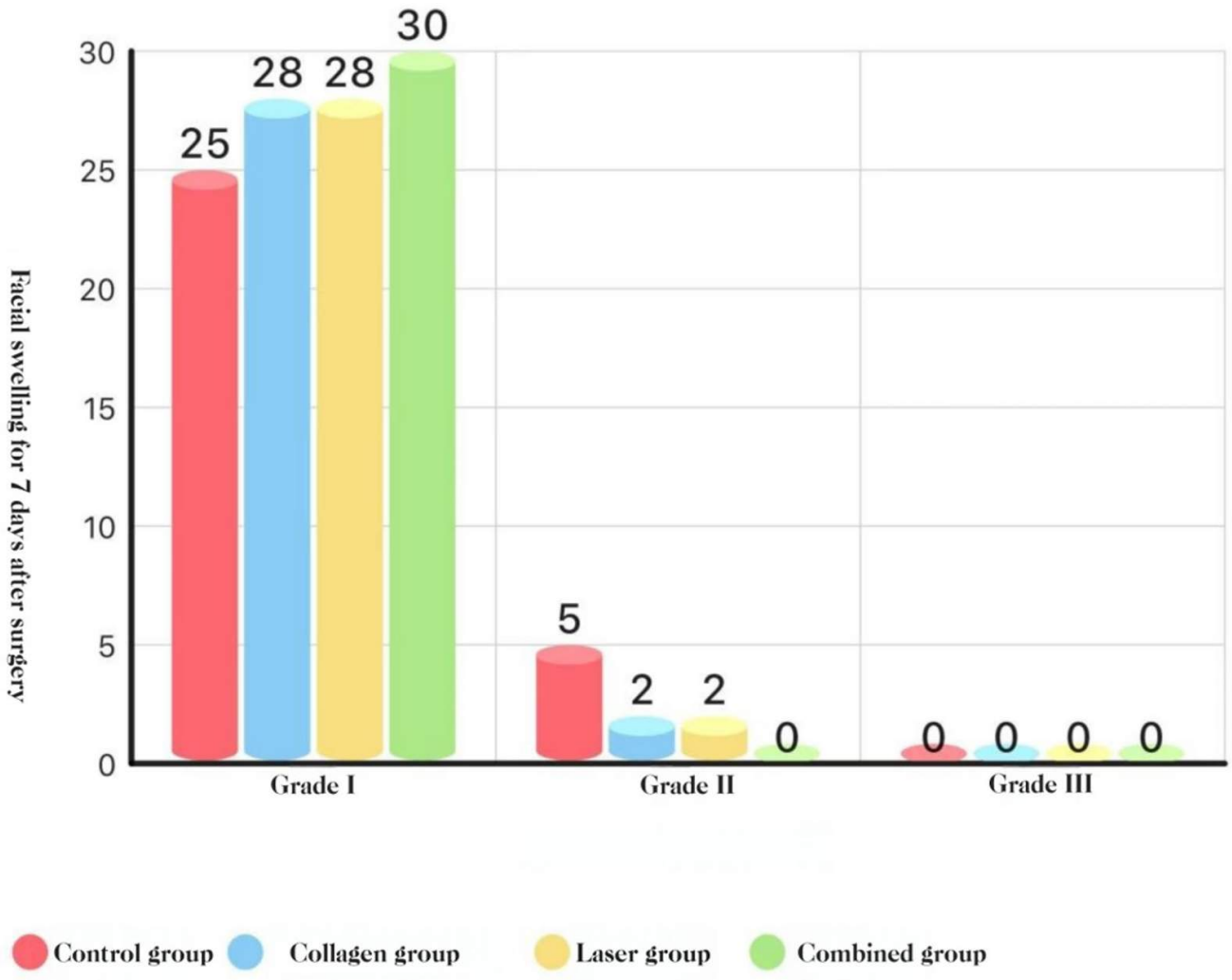


Fig. 10 Distribution of facial swelling at POD 7 (Marković–Todorović grades): histogram

Discussion

Observation of postoperative complications in cases treated with Er: YAG plus Nd: YAG lasers assisted by medical collagen

Pain

Our results indicate that both medical collagen alone and the combination of Er: YAG and Nd: YAG lasers reduce postoperative pain after extraction of impacted teeth, with the combined regimen providing superior analgesia. A meta-analysis of laser-assisted third-molar surgery reported that Er: YAG use improves postoperative pain [24], consistent with our findings. Because of its tissue penetration, the Nd: YAG laser can induce coagulation of cells—including neural elements—thereby decreasing nociceptor sensitivity and contributing to hemostasis and analgesia [20]. Bai Xijing and colleagues [25] similarly observed that application of medical collagen after tooth extraction reduces postoperative pain. In a split-mouth study of impacted mandibular third-molar removal, placement of a resorbable collagen sponge resulted in a significantly lower visual analogue scale (VAS) score at 1 week on the collagen side than on the control side ($P < 0.05$). Early postoperative facial swelling (weeks 1–2) was also reduced on the collagen side, although differences were no longer evident by week 14. These findings suggest that collagen sponge placement may lessen early postoperative complications [8], in line with our study.

Bleeding

At 30 min after extraction of impacted teeth, blood loss was lower in the collagen, laser, and combined-treatment groups than in the blank control group. This reduction is

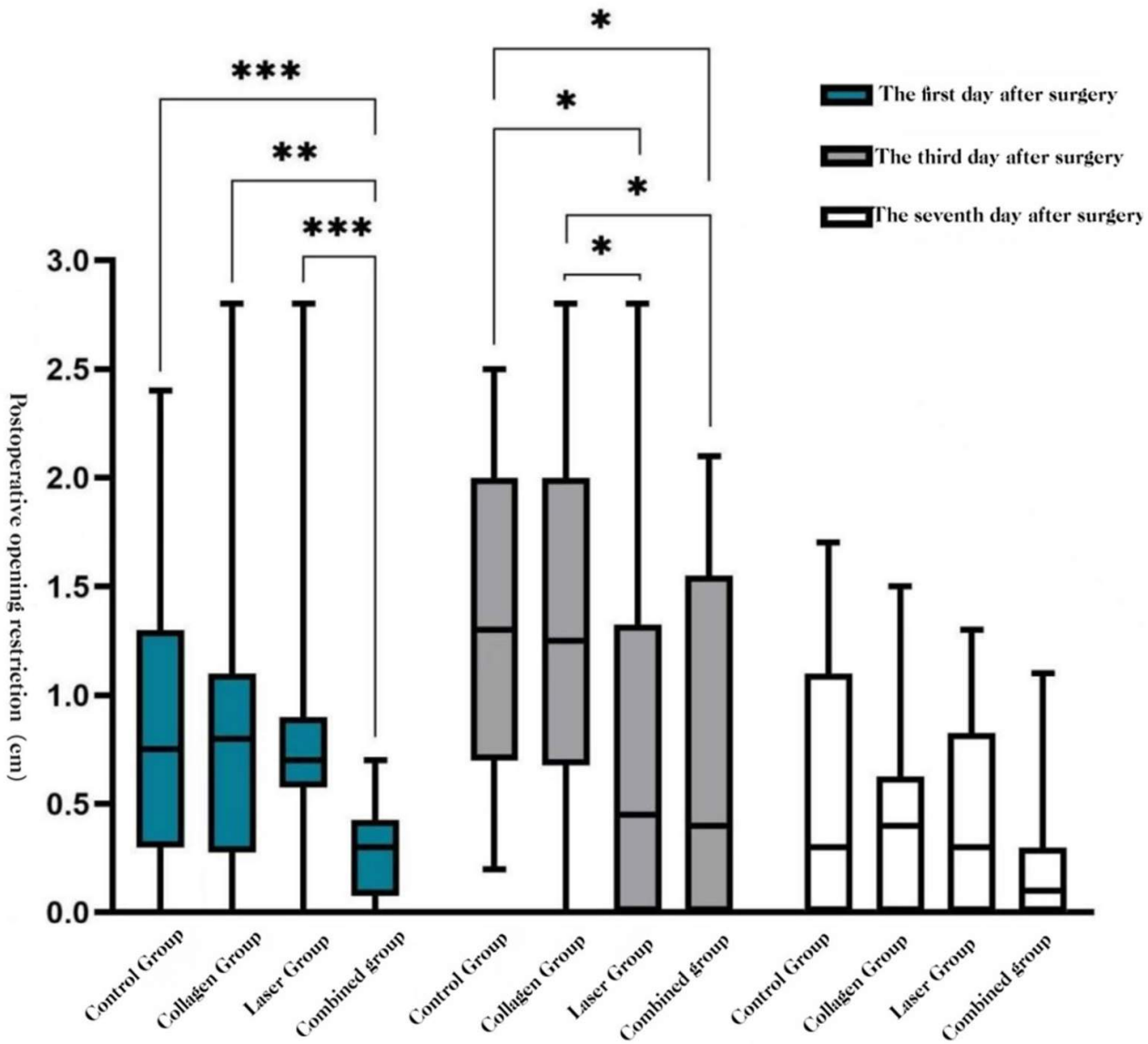


Fig. 11 Trismus: box-and-whisker plots by group (POD 1, 3, 7)

biologically plausible: the Nd: YAG laser, with relatively deep tissue penetration and melanin absorption, induces coagulation within the tissue and thereby promotes hemostasis [20]. Consistently, in a murine model, Nd: YAG irradiation produced instantaneous thrombus formation in dorsal skin, confirming its hemostatic action [26]. Collagen also contributes to hemostasis by swelling upon water absorption, stabilizing the clot and supporting wound closure [25]. Clinically, immediate application of a collagen-based cylindrical hemostatic dressing (ETIK COLLAGENE) after tooth extraction shortened mean bleeding time and reduced postoperative adverse events compared with an 8-minute delayed application [27]. Together, these data align with our findings of reduced early postoperative bleeding in the intervention groups.

Swelling

Our results show that both medical collagen alone and the combined use of Er: YAG and Nd: YAG lasers reduce postoperative swelling after removal of impacted third molars, with the combination providing the greatest benefit. Giovannacci et al. [15] reported less postoperative swelling with Er: YAG-assisted extraction compared with conventional turbine techniques, consistent with our findings. In a comparative study, postoperative Nd: YAG irradiation achieved

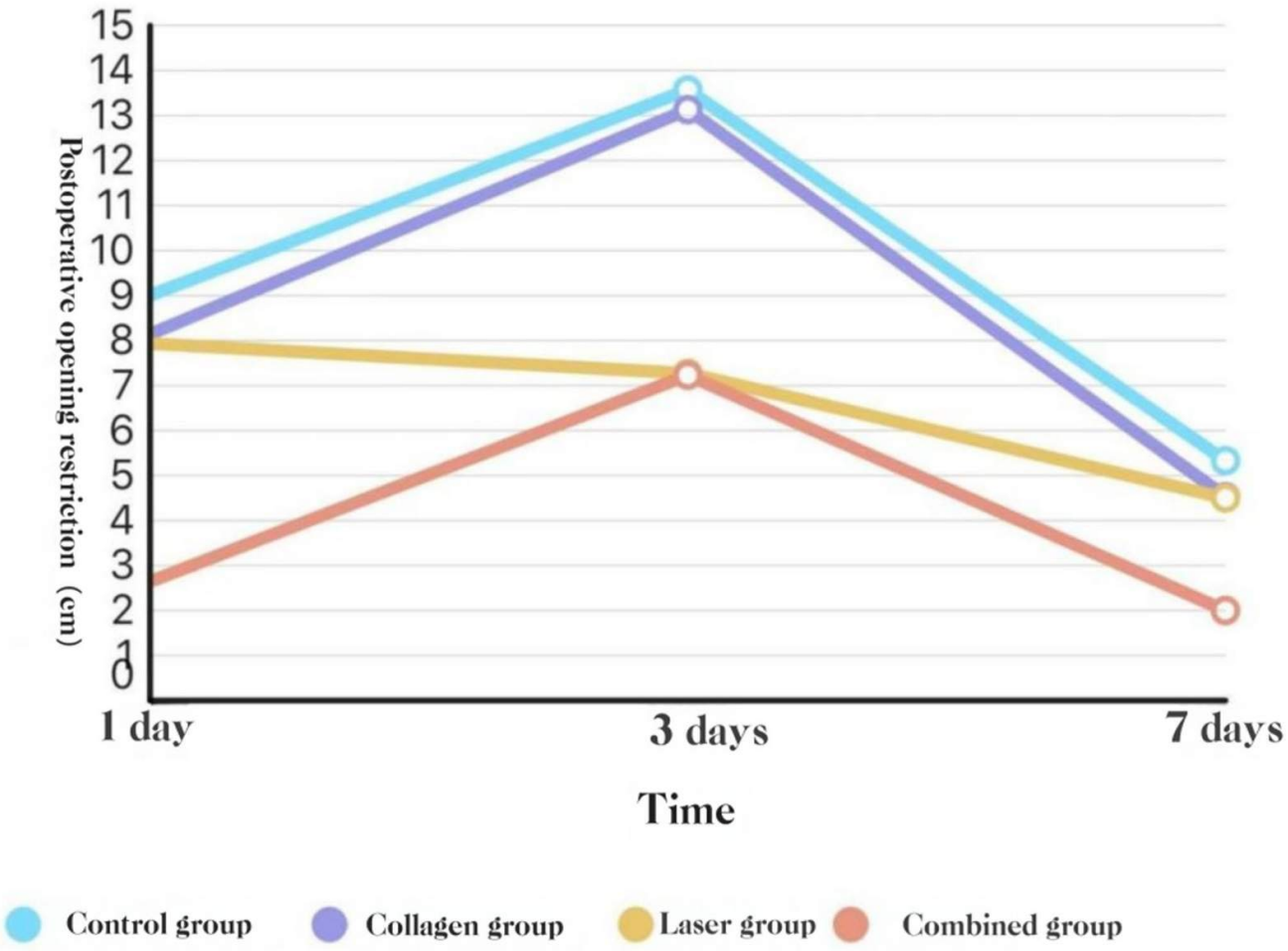


Fig. 12 Trismus over time by group: mean ΔMID (POD 1, 3, 7

swelling control comparable to intramuscular nonsteroidal anti-inflammatory drugs (NSAIDs), suggesting that Nd: YAG may limit reliance on NSAIDs without compromising efficacy. Jiao Kun and colleagues [28] likewise found that application of medical collagen after extraction attenuated postoperative swelling, in agreement with the present study.

Trismus

The combined use of Er: YAG and Nd: YAG lasers with adjunctive collagen reduced trismus on postoperative days 1 and 3. Trismus after extraction is largely driven by inflammation involving the masticatory muscles—the masseter, medial pterygoid, and temporalis—resulting in reactive spasm and limited mouth opening [5]. This observation is consistent with reports by Sales et al. [24] and Giovannacci [15]. Owing to its precise cutting and minimal collateral thermal injury, the Er: YAG laser can decrease surgical trauma during impacted-tooth surgery and thereby mitigate postoperative trismus. In addition, a study of non-surgical periodontal therapy showed that Er: YAG treatment improved probing depth and clinical attachment, findings that may translate to reduced postoperative trismus [29]. By contrast, placement of medical collagen into the socket after impacted-tooth extraction did not significantly affect trismus compared with blank controls, indicating that collagen alone does not alleviate postoperative limitation of mouth opening, in agreement with Jiao Kun et al. [28].

Cutaneous induration and ecchymosis

Post-extraction cutaneous induration and ecchymosis stem from two processes: (i) trauma-related bleeding with intratissue blood stasis and (ii) postoperative inflammation that increases capillary permeability, impairs lymphatic drainage, and sustains edema; when unresolved, these changes manifest as induration and bruising. Our results show that the combined-treatment group had fewer cases of cutaneous induration than either the collagen or laser group, and both had fewer cases than the blank control group; these

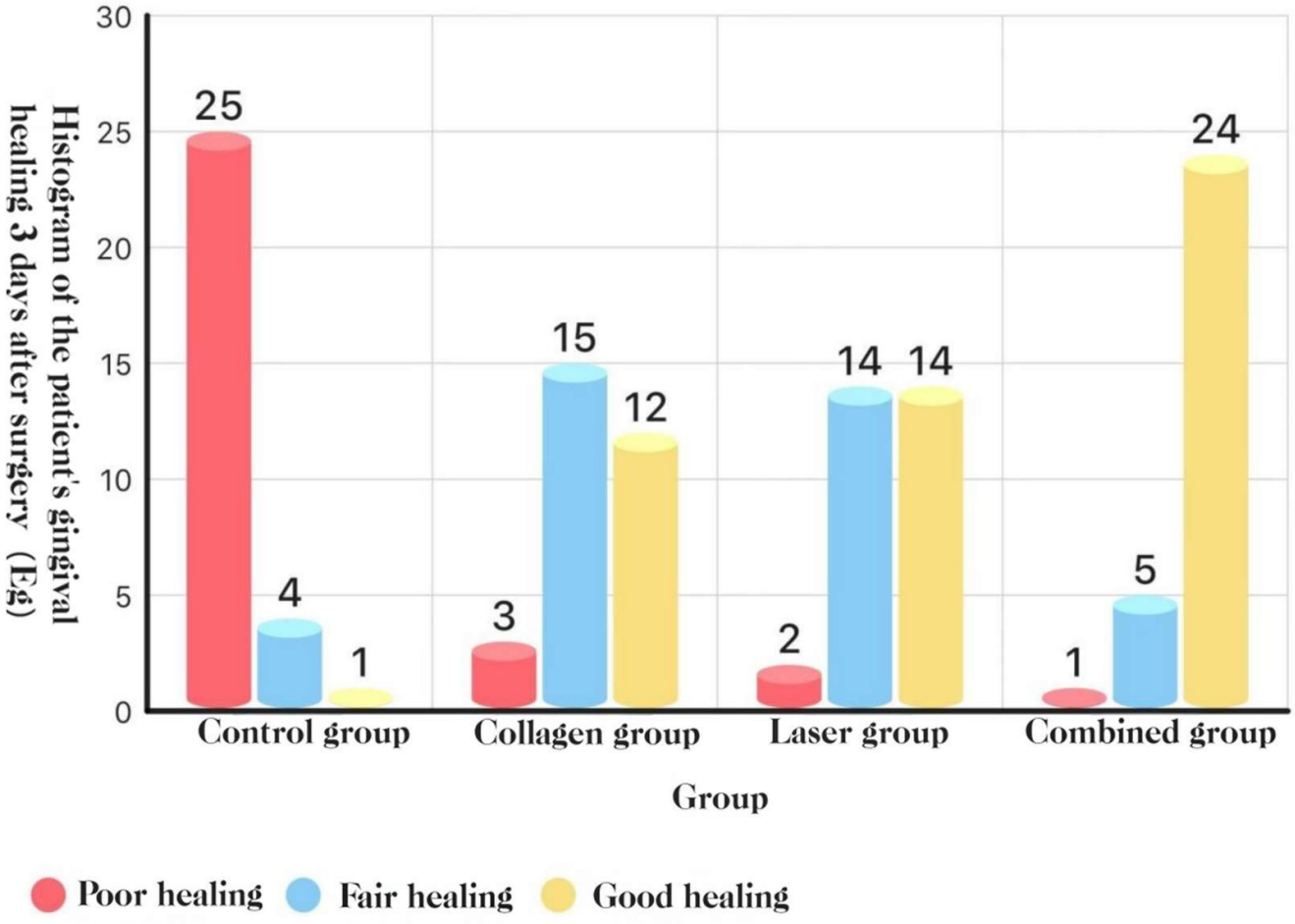


Fig. 13 Distribution of mucosal healing grades at POD 3: histogram

differences were statistically significant. These findings suggest that collagen, Nd: YAG therapy, and their combination reduce post-extraction cutaneous induration after removal of impacted teeth. By contrast, the incidence of cutaneous ecchymosis did not differ significantly among the four groups; however, absolute counts favored the combination (2 cases) over the blank control (10 cases). Although not statistically significant, this trend may still be clinically relevant. The results are consistent with the hematoma outcomes reported by Giovannacci et al. [15]. Given the pathophysiology of hematoma and ecchymosis, minimizing bleeding remains a key preventive strategy; as discussed above, collagen, Er: YAG plus Nd: YAG therapy, or their combination can reduce bleeding and thereby lower the risk of cutaneous induration and ecchymosis.

Alveolar osteitis

No cases of alveolar osteitis occurred in any of the four groups. Nd: YAG laser therapy may facilitate post-extraction wound healing in patients at high risk for jaw osteomyelitis, such as those receiving long-term bisphosphonates [30]. Nd: YAG irradiation also promotes osteogenesis by upregulating BMP-2 expression in pre-osteoblasts, thereby inducing cellular proliferation and differentiation; when applied within periodontal pockets, it exerts additional antibacterial effects [31]. Bai Xijing et al. [25] reported that medical collagen enhances hemostasis after removal of impacted teeth, reduces postoperative pain, and lowers the incidence of alveolar osteitis. Similarly, Yuan Xuguang et al. [32] found in a large comparative cohort that collagen substantially decreases the risk of alveolar osteitis. In light of our results, plausible explanations include: (i) advances toward minimally invasive extraction—using high-speed contra-angle electric handpieces and piezosurgery rather than traditional techniques—which markedly reduce the risk of alveolar osteitis; and (ii) standardized operative protocols, strong patient adherence, and adjuvant use of antibiotics and mouth rinses.

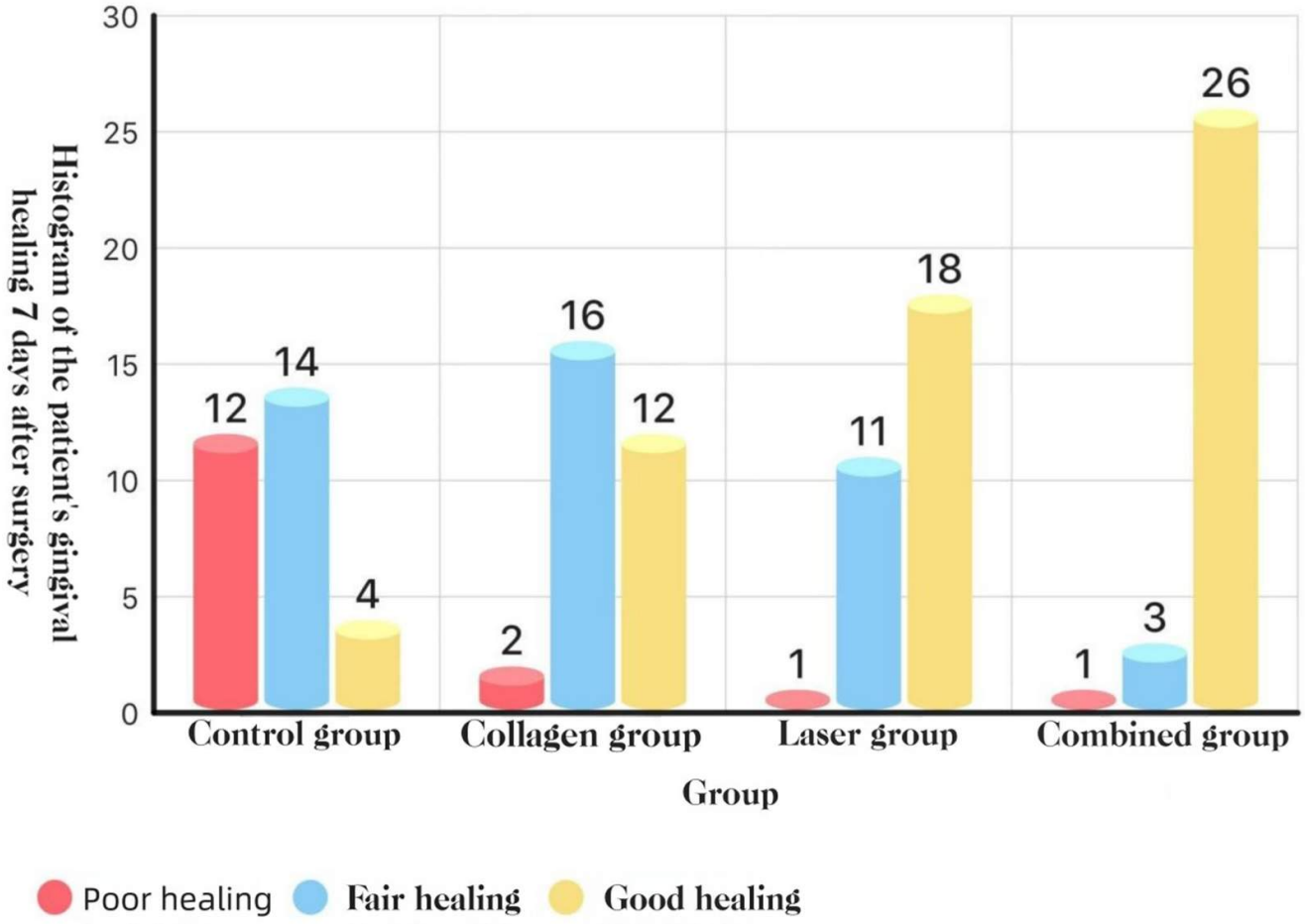


Fig. 14 Distribution of mucosal healing grades at POD 7: histogram

Mucosal healing

Postoperative assessments showed higher proportions of "good" mucosal healing on days 3 and 7 in the combination, collagen, and laser groups than in the blank control group, with statistically significant differences. Moreover, the combination group outperformed both single-modality groups, indicating that the combined regimen better supports mucosal healing after removal of impacted teeth. Mechanistically, Er: YAG irradiation promotes type I collagen formation and osteogenesis and, together with the biostimulatory and antimicrobial actions of laser therapy, facilitates wound repair [15, 16]. Nd: YAG irradiation further inactivates bacteria and virulence factors such as endotoxin, thereby reducing postoperative infection and promoting soft-tissue healing [19]. In patients on long-term bisphosphonates, Nd: YAG use enhanced gingival and bone healing after extraction [30]. Consistently, Ma Yang et al. [33] reported that medical collagen promotes soft-tissue repair. As a biocompatible scaffold, collagen— the principal organic component of bone—supports bone formation and mineralization. Collagen sponges promote post-extraction bone healing through their porous architecture, which favors cell seeding and osteoblastic differentiation [34]. In impacted third-molar surgery, collagen sponges are placed in extraction sockets to prevent soft-tissue collapse, reduce postoperative complications, and promote early healing of soft tissues and periodontal defects [35]. During alveolar-socket healing, collagen induces a vascular network rich in H-type vessels, fostering new bone formation critical to early socket repair [36]. In summary, collagen sponges and membranes enhance bone repair and reduce complications, making them key materials for alveolar-ridge preservation; future studies should clarify their long-term effects on bone healing and periodontal regeneration.

Pro-inflammatory cytokines: IL-1β, IL-6, and TNF-α

Interleukin-1β (IL-1β) and interleukin-6 (IL-6) are chemotactic cytokines that amplify inflammation, activate neutrophils and macrophages, promote alveolar bone resorption and febrile responses, and play key roles in infection and

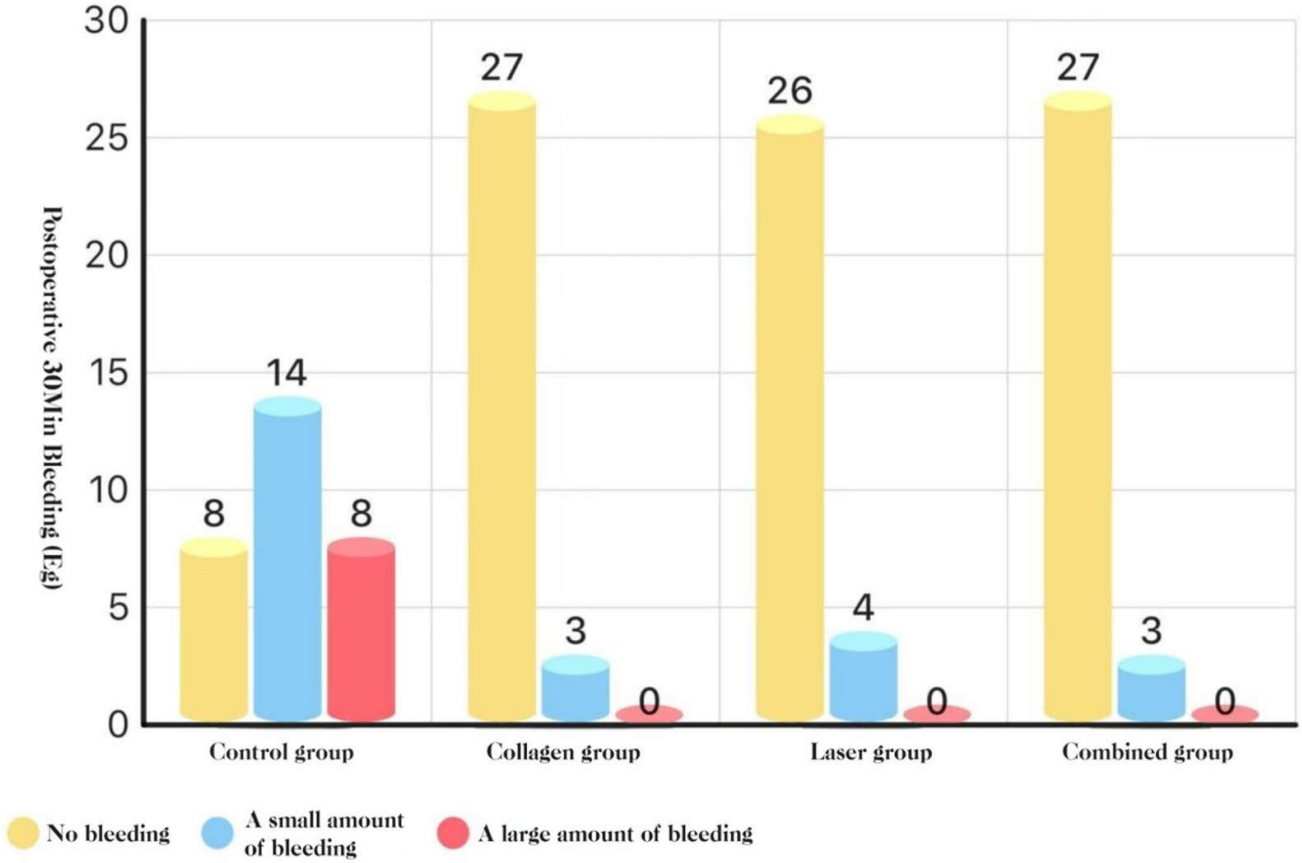


Fig. 15 Distribution of bleeding grades at 30 minutes post-extraction: histogram

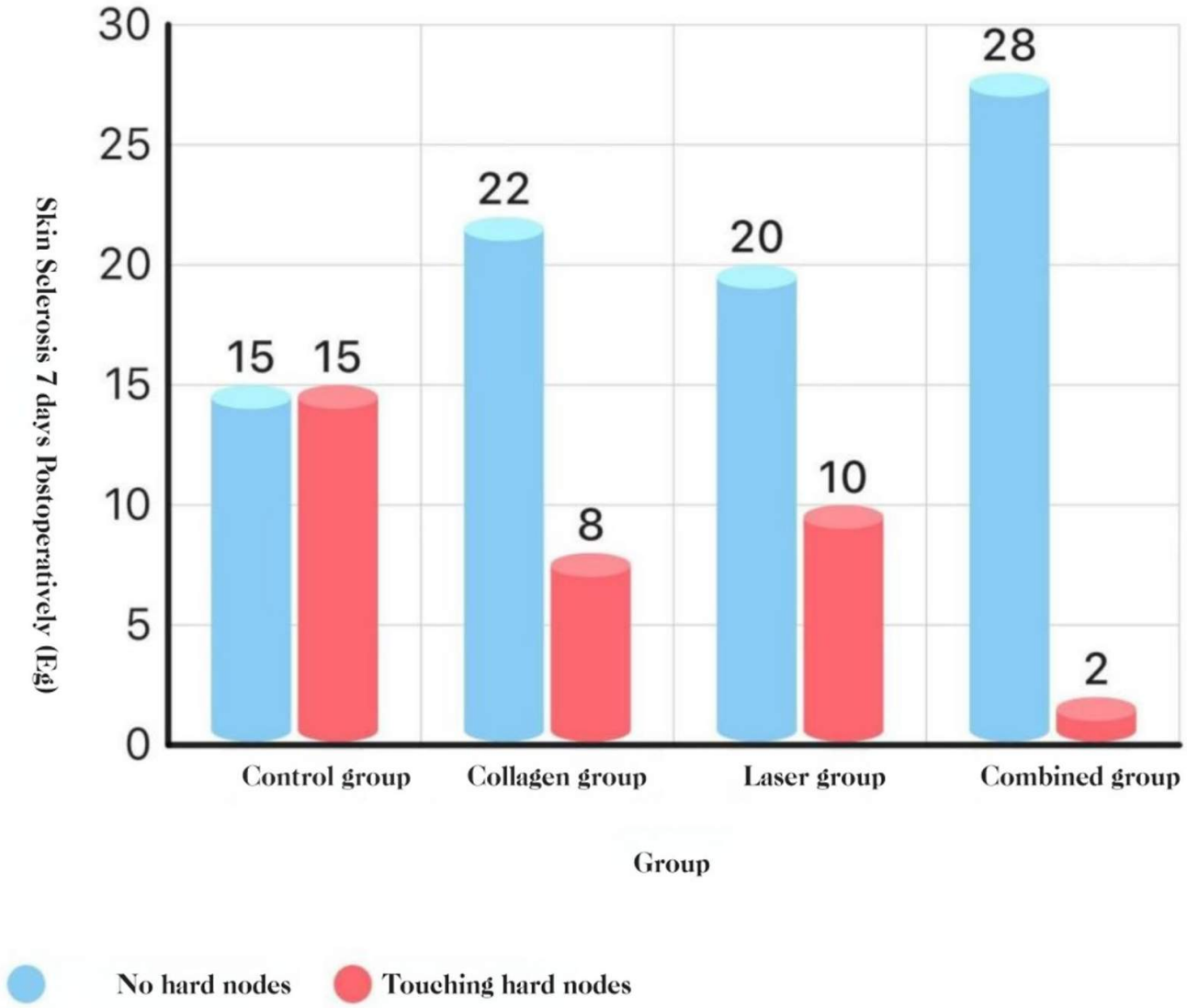


Fig. 16 Cutaneous induration at POD 7

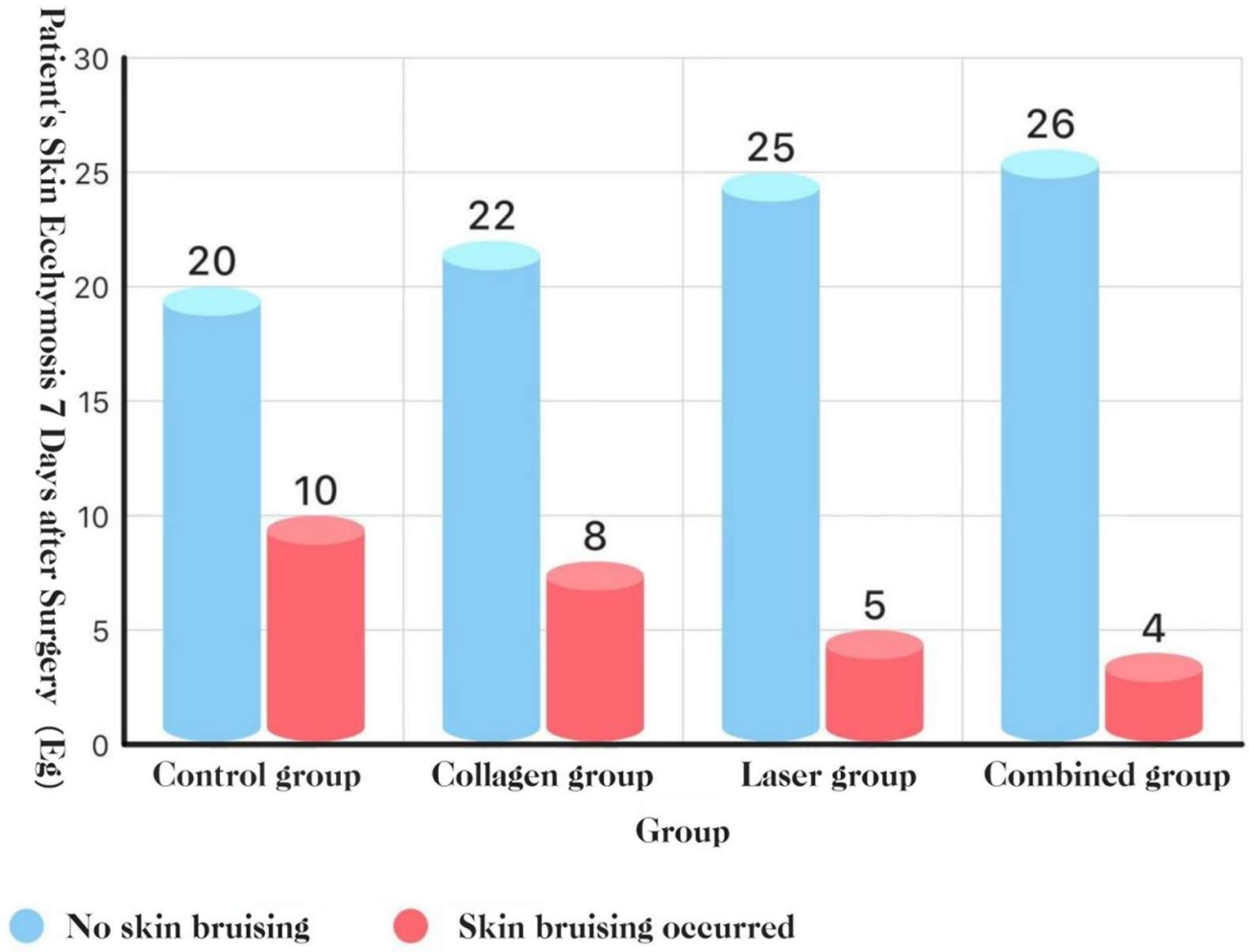


Fig. 17 Cutaneous ecchymosis at POD 7

tissue injury [37]. Tumor necrosis factor-α (TNF-α) is a pivotal early mediator that recruits inflammatory cells to sites of infection; it activates neutrophils and lymphocytes, modulates tissue metabolism, and stimulates the release of other cytokines (e.g., IL-6 and IL-1β), acting synergistically in pain signaling [38].

In this study, local cytokine production was evident after extraction of impacted teeth; however, combining Er: YAG and Nd: YAG lasers with adjunctive collagen reduced IL-1β, IL-6, and TNF-α levels at the surgical site, which may partially account for the concomitant reduction in complications. Guo Tao et al. [39] reported that Er: YAG laser debridement of inflammatory granulation tissue accelerated healing and decreased postoperative pain and inflammation, with lower TNF-α, IL-1β, and IL-6 concentrations in socket exudate than in conventional care—findings consistent with ours. Similarly, Bai Xijing et al. [25] observed that placement of collagen within the extraction socket lowered inflammatory cytokine levels, again aligning with the present results.

Overall, all three interventions—combination therapy, laser alone, and collagen alone—outperformed the blank control in mitigating postoperative complications. Among them, the combination therapy yielded the greatest reductions in pain, bleeding, facial swelling, trismus, and cutaneous induration after removal of impacted teeth, followed by the Er: YAG/Nd: YAG laser regimen and then collagen alone.

Conclusion

The combined application of Er: YAG and Nd: YAG lasers with adjunctive collagen in impacted third-molar surgery reduces postoperative complications—including pain, bleeding, facial swelling, trismus, and cutaneous

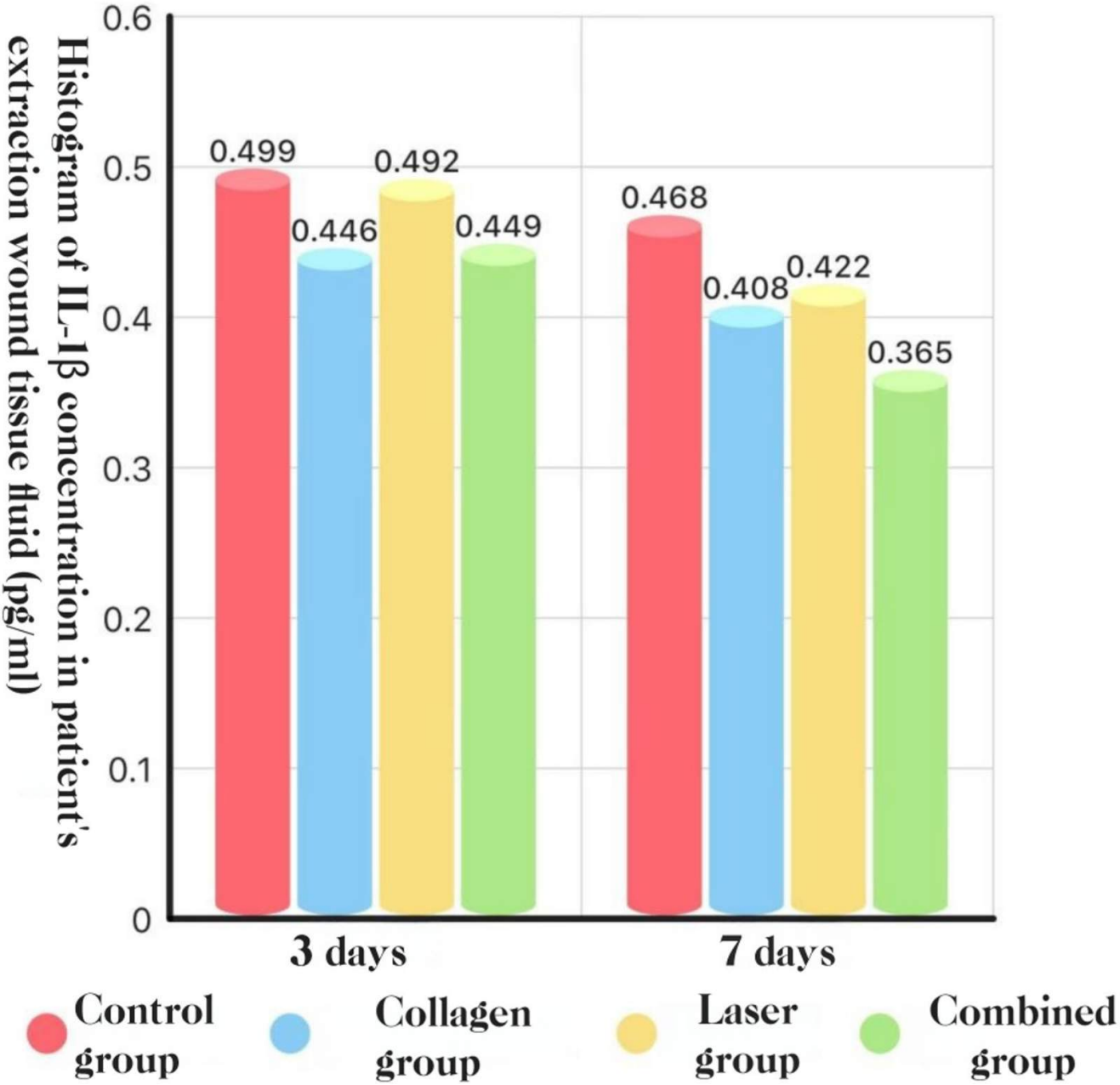


Fig. 18 IL-1β concentration in socket exudate at POD 3 and POD 7: histogram

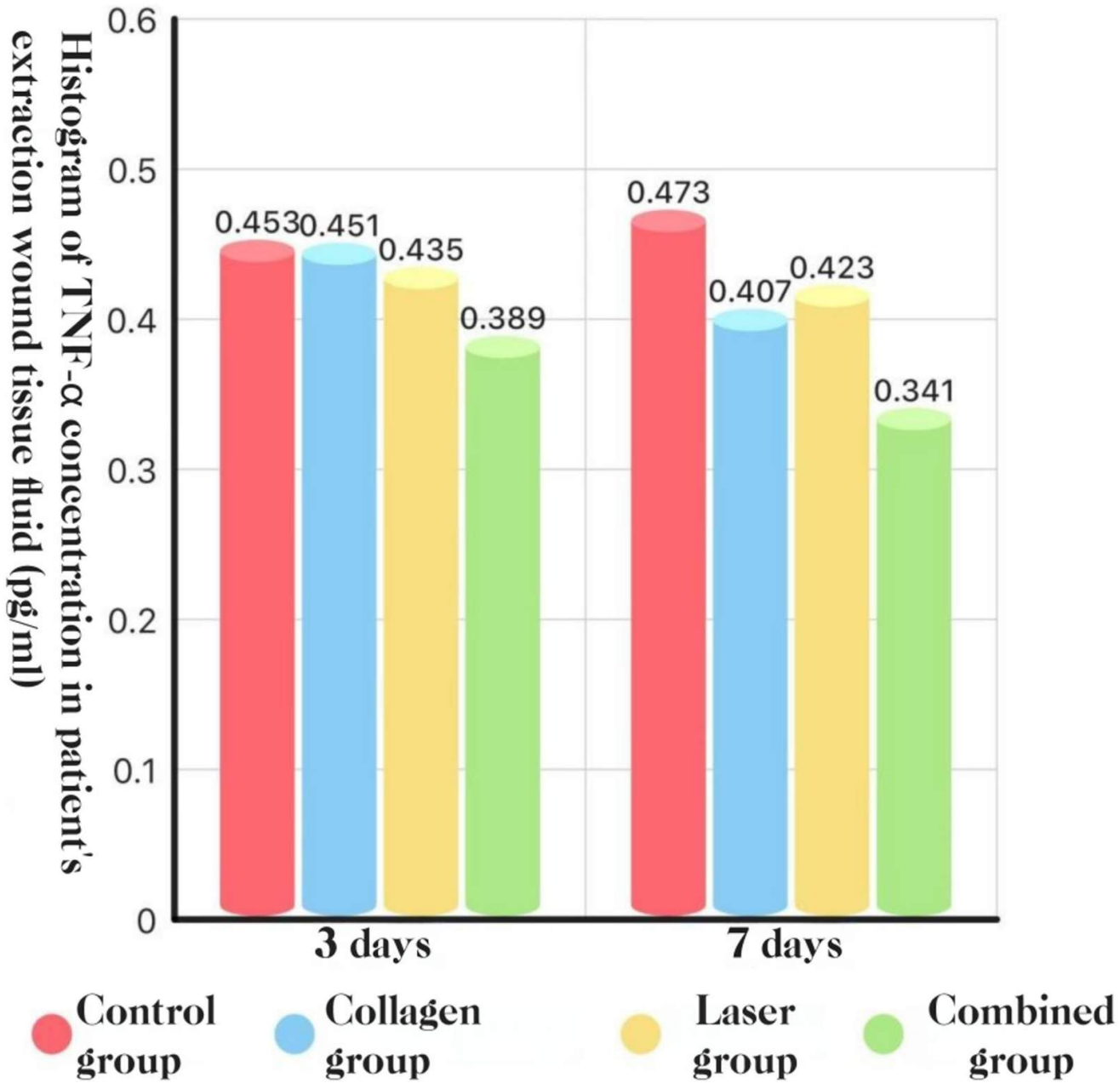


Fig. 19 TNF-α concentration in socket exudate at POD 3 and POD 7: histogram

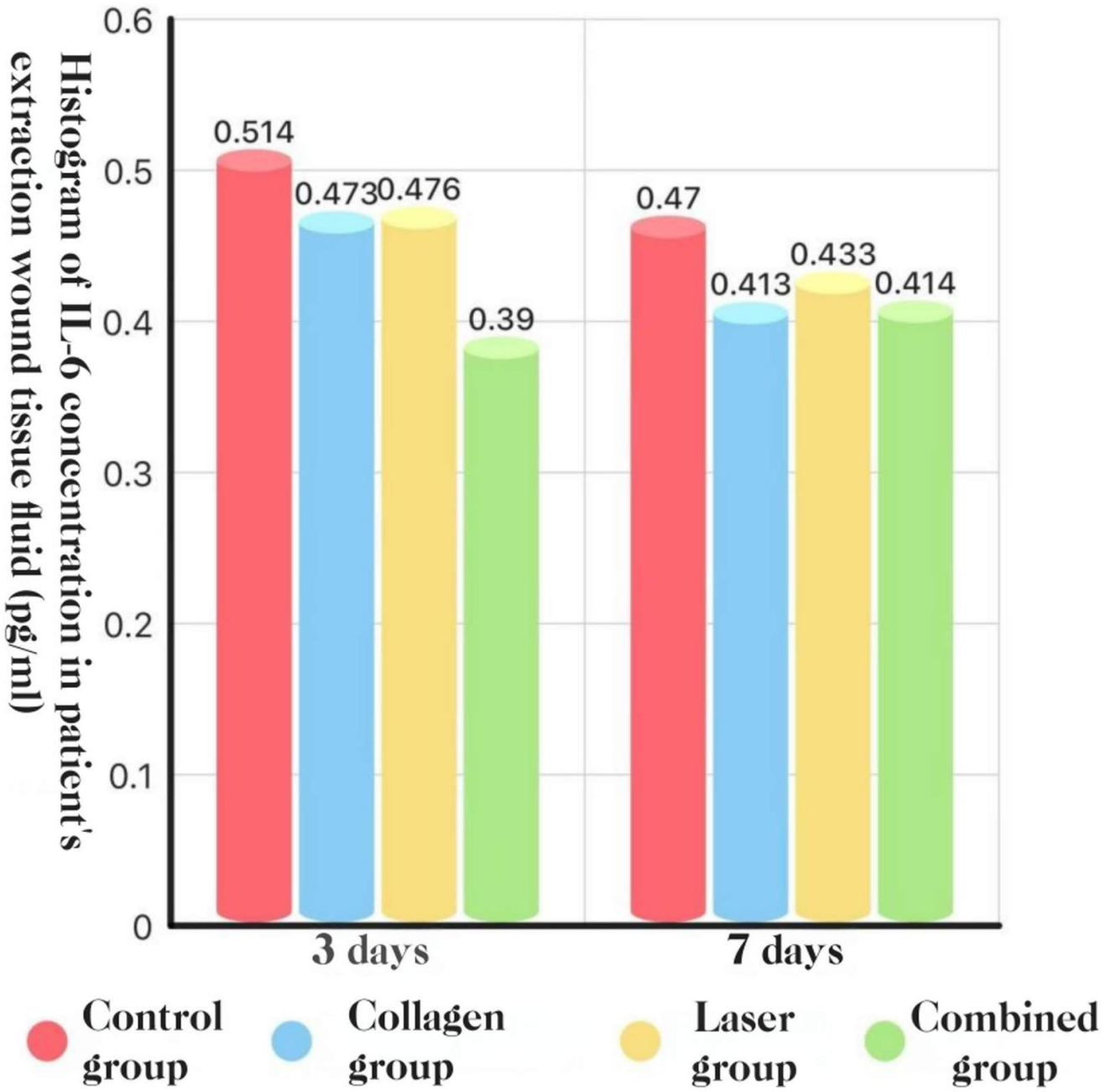


Fig. 20 IL-6 concentration in socket exudate at POD 3 and POD 7: histogram



Fig. 21 A, almost no obvious facial swelling at postoperative day 3; B, mucosal healing at postoperative day 3; C, mucosal healing at postoperative day 7; D, mucosal healing at postoperative week 3

induration—facilitates socket healing, and improves postoperative quality of life, supporting broader clinical adoption. Moreover, this regimen lowers local levels of TNF-α, IL-1β, and IL-6 at the surgical site, offering clinically relevant evidence to inform practice.

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Author contributions Tonghan Zhang guided the whole experiment. Zhantang Yuan and Zhihui Xu were responsible for the implementation of the experiment and the writing of the thesis.Xia Wan was responsible for organising the experimental data.All authors reviewed the manuscript.

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Data availability Additional data are available in the Supplementary Material.

Declarations

Competing interests The authors declare no competing interests.

Clinical trial number PID:248,812.

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References

1. Zhang Z (2020) Oral and maxillofacial surgery, 8th edn. People's Medical Publishing House, Beijing, China. [in Chinese]
2. Ghaeminia H, Nienhuijs ME, Toedtling V et al (2020) Surgical removal versus retention for the management of asymptomatic disease-free impacted wisdom teeth. Cochrane Database Syst Rev 5(5):CD003879
3. Bailey E, Kashbour W, Shah N, Worthington HV, Renton TF, Coulthard P (2020) Surgical techniques for the removal of mandibular wisdom teeth. Cochrane Database Syst Rev 7(7):CD004345
4. KalaiSelvan S, Ganesh SKN, Natesh P, Moorthy MS, Niazi TM, Babu SS (2020) Prevalence and pattern of impacted mandibular third molar: an institution-based retrospective study. J Pharm Bioallied Sci 12(Suppl 1):S462–S467
5. Dignam P, Elshafey M, Jeganathan A, Foo M, Park JS, Ratnaweera M (2024) Prevalence and factors influencing post-operative complications following tooth extraction: a narrative review. Int J Dent 2024:7712829
6. Sayed N, Bakathir A, Pasha M et al (2019) Complications of third molar extraction: a retrospective study from a tertiary healthcare centre in Oman. Sultan Qaboos Univ Med J 19(3):e230–e235
7. Kokash M, Darwich K, Ataya J (2023) The effect of hyaluronic acid addition to collagen in reducing trismus and swelling after surgical extraction of impacted lower third molars: a split-mouth, randomized controlled study. Clin Oral Investig 27(8):4659–4666
8. Kim JW, Seong TW, Cho S, Kim SJ (2020) Randomized controlled trial on the effectiveness of absorbable collagen sponge after extraction of impacted mandibular third molar: split-mouth design. BMC Oral Health 20(1):77
9. Manafikhi M, Ataya J, Heshmeh O (2022) Evaluation of the efficacy of platelet-rich fibrin (I-PRF) intra-articular injections in the management of internal derangements of temporomandibular joints: a controlled preliminary prospective clinical study. BMC Musculoskelet Disord 23(1):454
10. Ng E, Tay JRH, Ong MMA (2021) Minimally invasive periodontology: a treatment philosophy and suggested approach. Int J Dent 2021:2810264
11. Cinquini C, Izzetti R, Nisi M, Gulia F, Garcia Mira B, Barone A (2025 Jul) *Limosilactobacillus reuteri* improves healing following fully impacted tooth extractions: randomized clinical trial. Oral Dis 13. https://doi.org/10.1111/odi.70026Epub ahead of print
12. Stuermer EK, Bang C, Giessler A et al (2024) Effect of oral multispecies probiotic on wound healing, periodontitis and quality of life in patients with diabetes. J Wound Care 33(6):394–407
13. Tsai SJ, Chen MH, Lin HY, Lin CP, Chang HH (2019) Pure type-I collagen application to third molar extraction socket reduces postoperative pain score and duration and promotes socket bone healing. J Formos Med Assoc 118(1 Pt 3):481–487
14. Paghdiwala AF, Vaidyanathan TK, Paghdiwala MF (1993) Evaluation of erbium:yag laser radiation of hard dental tissues: analysis of temperature changes, depth of cuts, and structural effects. Scanning Microsc 7(3):989–997
15. Giovannacci I, Giunta G, Pedrazzi G et al (2018) Erbium yttrium-aluminum-garnet laser versus traditional bur in the extraction of impacted mandibular third molars: analysis of intra- and postoperative differences. J Craniofac Surg 29(8):2282–2286
16. Aoki A, Mizutani K, Taniguchi Y et al (2024) Current status of Er:YAG laser in periodontal surgery. Jpn Dent Sci Rev 60:1–14
17. Hadian K, Babossalam S, Mahdikia H et al (2022) Efficacy and safety of non-thermal nitrogen plasma versus long-pulsed nd:yag laser for hand rejuvenation. Lasers Med Sci 37:181–191
18. Grzech-Leśniak K, Belvin BR, Lewis JP, Golob Deeb J (2021) Treatment with nd:yag laser irradiation combined with sodium hypochlorite or hydrogen peroxide irrigation on periodontal pathogens: an in vitro study. Photobiomodul Photomed Laser Surg 39(1):46–52
19. Aoki A, Mizutani K, Schwarz F et al (2015) Periodontal and peri-implant wound healing following laser therapy. Periodontol 2000 68:217–269
20. Han Y, Zhu J, Hu S, Li C, Zhang X (2023) Nd:YAG laser therapy on postoperative pain, swelling, and trismus after mandibular third molar surgery: a randomized double-blinded clinical study. Lasers Med Sci 38(1):176
21. Markovic A, Todorovic L (2007) Effectiveness of dexamethasone and low-power laser in minimizing oedema after third molar surgery: a clinical trial. Int J Oral Maxillofac Surg 36(3):226–229
22. Marini L, Rojas MA, Sahrmann P, Aghazada R, Pilloni A (2018) Early wound healing score: a system to evaluate the early healing

of periodontal soft tissue wounds. J Periodontal Implant Sci 48(5):274–283
23. Pippi R (2017) Post-surgical clinical monitoring of soft tissue wound healing in periodontal and implant surgery. Int J Med Sci 14(8):721–728
24. Sales PHDH, Barros AWP, Silva PGB, Vescovi P, Leão JC (2022) Is the er:yag laser effective in reducing pain, edema, and trismus after removal of impacted mandibular third molars? A meta-analysis. J Oral Maxillofac Surg 80(3):501–516
25. Bai X et al (2019) Effect analysis of collagen sponge used in mandibular impacted tooth extraction. Chin J Oral Maxillofacial Surg 17(2):155–158 [in Chinese]
26. Li D, Wu W, Chen B, Gou Z, Zhao P, Wang G (2020) Thermal coagulum formation and hemostasis during repeated multipulse Nd:YAG laser treatment of cutaneous vascular lesions: animal experiment study. Lasers Med Sci 35(7):1589–1597
27. Gu JT, Jiao K, Li J et al (2021) Polyphosphate-crosslinked collagen scaffolds for hemostasis and alveolar bone regeneration after tooth extraction. Bioact Mater 15:68–81
28. Jiao K, Sun B et al (2020) Effect analysis of Batroxobin and collagen sponge in mandibular impacted tooth extraction. Chin J Experimental Diagnostics 24(7):1095–1099 [in Chinese]
29. Aoki A, Mizutani K, Mikami R et al (2023) Er:YAG laser-assisted comprehensive periodontal pocket therapy for residual periodontal pocket treatment: a randomized controlled clinical trial. J Periodontol 94(10):1187–1199
30. Vescovi P, Giovannacci I, Merigo E et al (2015) Tooth extractions in high-risk patients under bisphosphonate therapy and previously affected with osteonecrosis of the jaws: surgical protocol supported by low-level laser therapy. J Craniofac Surg 26(3):696–699
31. Choi AY, Reddy CM, McGary RT et al (2019) Adjunctive Nd:YAG laser irradiation for ridge preservation and immediate implant procedures: a consecutive case series. Clin Adv Periodontics 9(3):125–134
32. Yuan X, Li T, Zhu J et al (2019) Clinical application of medical collagen sponge in preventing dry socket after extraction of mandibular impacted third molars. J Clin Stomatology 35(2):92–94 [in Chinese]
33. Ma Y, Zhou H, Zhang Z et al (2021) New materials for tooth extraction. Chin J Practical Stomatology 14(6):660–663 [in Chinese]
34. Wang L, Li W, Qu Y et al (2022) Preparation of super-absorbent and highly active fish collagen sponge and its hemostatic effect in vivo and in vitro. Front Bioeng Biotechnol 10:862532
35. Moreno-Drada JA, Abreu LG, Lino PA, Parreiras Martins MA, Pordeus IA, de Nogueira Guimarães Abreu MH (2021) Effectiveness of local hemostatics to prevent bleeding in dental patients on anticoagulation: a systematic review and network meta-analysis. J Craniomaxillofac Surg 49:570–583
36. Wu MH, Shih MH, Hsu WB et al (2017) Evaluation of a novel biodegradable thermosensitive keto-hydrogel for improving postoperative pain in a rat model. PLoS One 12(10):e0186784
37. Pan Q, Wang P (2019) Influence of the regulatory relationship between miR-202 and inflammatory cytokines on the pathologic development of chronic periodontitis. Mod J Stomatol 33(4):223–228
38. Polak D, Shapira L (2018) An update on the evidence for pathogenic mechanisms that may link periodontitis and diabetes. J Clin Periodontol 45(2):150–166
39. Guo T, Yang T, Xu Y et al (2021) Evaluation of minimally invasive extraction combined with er:yag laser wound management for impacted teeth. J Xinjiang Med Univ 44(11):1270–1274 [in Chinese]

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